

Magy Progress Report

Project title: Magy: Designing an Immersive, Interactive Platform for Gamified Instruction and Discovery
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1. Executive Summary

Magy has moved from the original proposal stage into a working prototype with two concrete subsystems:

1. A course generation system that transforms source course materials into structured learning assets.
2. A learner-facing course learning system that renders those assets as an interactive, gamified study experience.

The current implementation is built around a real sample course, COMP7415A Algorithmic Trading. The system now contains a ready course catalog, generated lesson routes, step-level question banks, textbook-style MDX materials, relevance reports, validation commands, and two browser-based frontends for learner delivery and creator review.

The prototype is not yet a production deployment, but it already demonstrates the main technical direction of the project: generated educational content is not kept as loose text. It is converted into structured, validated, reviewable, and playable learning units.

2. Current System Status

The current repository contains one indexed course package:

Item	Current status
Ready courses in catalog	1
Course chapters	11
Plain-text lesson conversions	11
Generated guided-story steps	75
Step-local question banks	75
Question families	686
Question variants	1,445
Generated textbook MDX files	11
Catalog validation result	ready=1, blocked=0

The generated course covers the full lecture sequence from course introduction to machine learning in algorithmic trading. Each lecture is represented as a chapter with a guided story route, practice assets, and a

textbook file.

3. Course Generation System

The generation pipeline has been implemented as a Rust CLI called `coursegen`. It currently supports the following workflow:

1. Convert raw lecture PDFs into reusable plain text.
2. Generate guided-story lesson steps from the plain text and course context.
3. Store each generated step as a self-contained `step.json`.
4. Generate a colocated `question_bank.json` for each step.
5. Generate a lesson-level textbook in MDX.
6. Score generated assets against course and exam signals where available.
7. Validate the catalog before exposing generated courses to the player.

This staged design is important because a course package contains several related artifact families. A single prompt-response output would be hard to inspect, validate, or resume. The current layout instead gives each step its own content and question bank, while the course index remains the single catalog source for the frontend.

The backend also now includes:

- reusable stage execution, so existing valid assets can be reused instead of regenerated;
- targeted regeneration controls through `--force-stage`;
- a DeepSeek API provider option for large-context model calls;
- cached exam text conversion for relevance scoring;
- backend-owned catalog validation that marks a course as `ready` or `blocked`.

This directly supports the project objective of making AI-assisted content generation more controllable and auditable.

4. Learner-Facing Course Learning System

The learner app is a React/Vite prototype that reads the generated course catalog and pipeline assets directly. It presents the generated course as a learning game rather than as a file browser.

Current learner-facing functions include:

- course library and course home views;
- chapter route selection;
- immersive guided-story playback;
- interactive checkpoint questions inside story screens;
- local progress tracking;
- textbook reading mode from generated MDX;
- flashcard-style recall session;
- practice vault separated into flashcards, quizzes, and longform items;
- backend validation awareness, so invalid courses can be blocked before students enter them.

This implements the core learner-delivery part of the proposal. The current interaction model is not just static reading: the student enters a route, advances through scenes, answers checkpoints, reviews cards, and explores a practice bank.

Learner System Screenshots

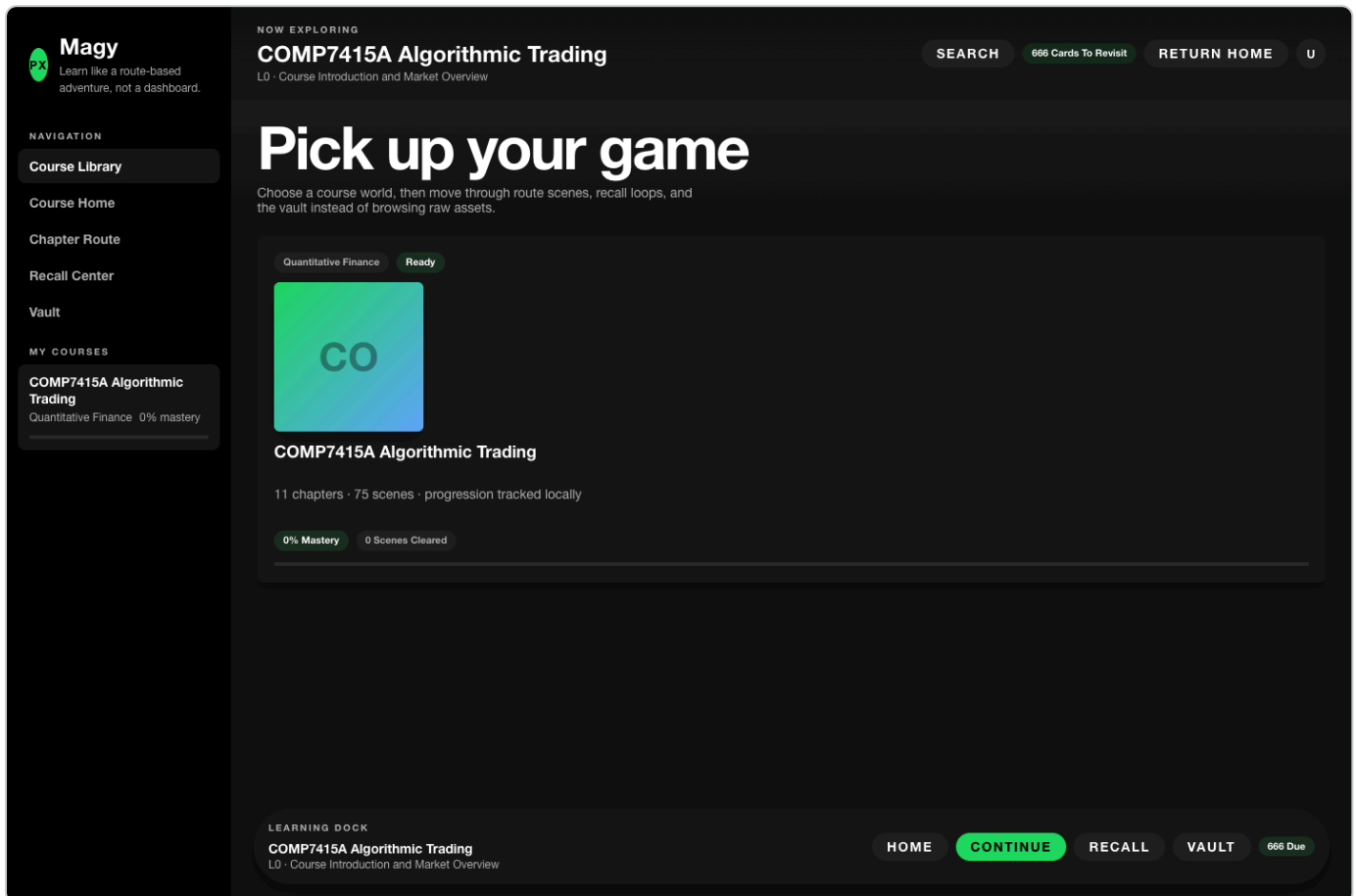


Figure 1. Magy learner library showing a ready generated course package.

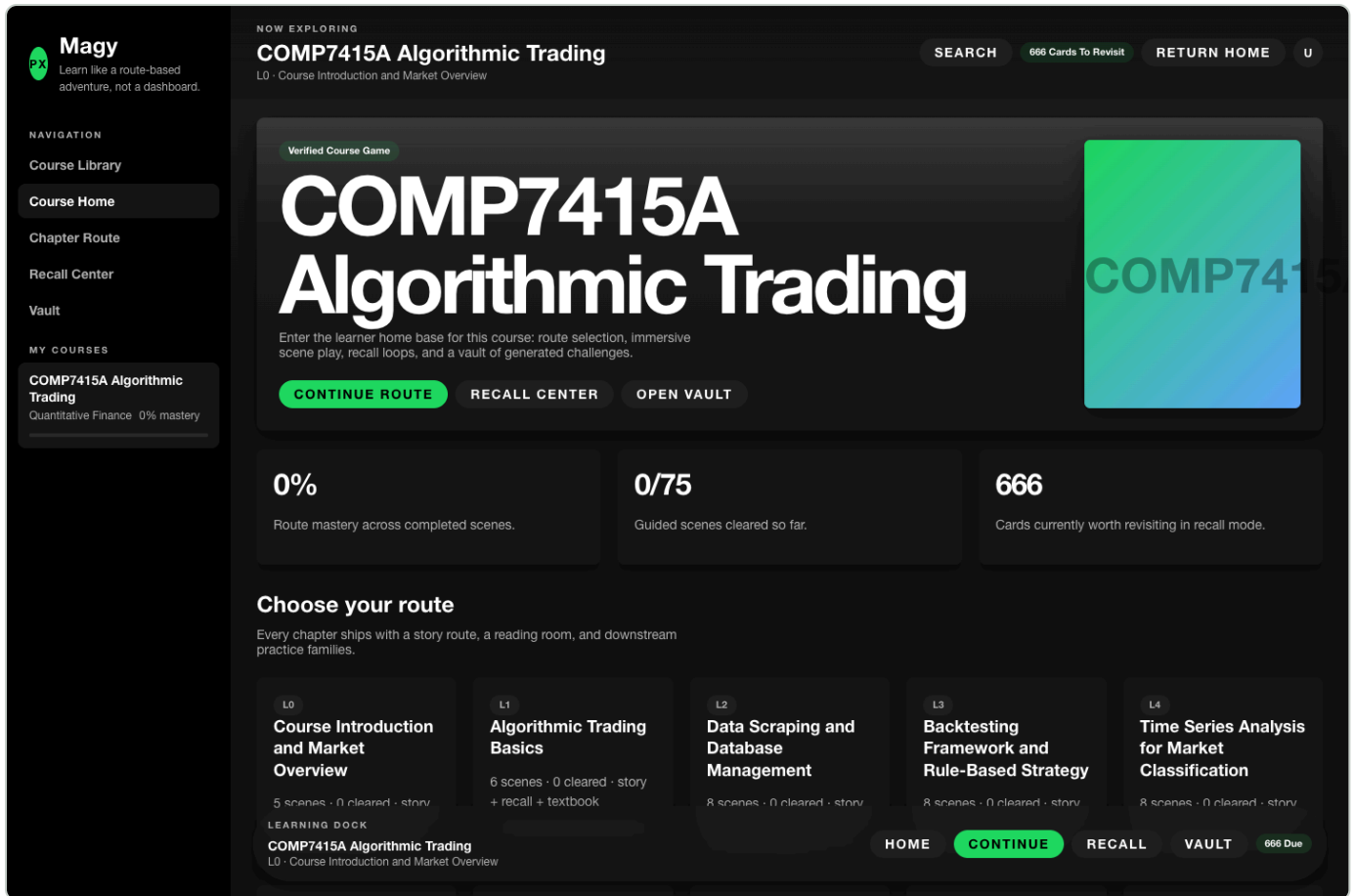


Figure 2. Course home with route progress, recall entry, and vault entry.

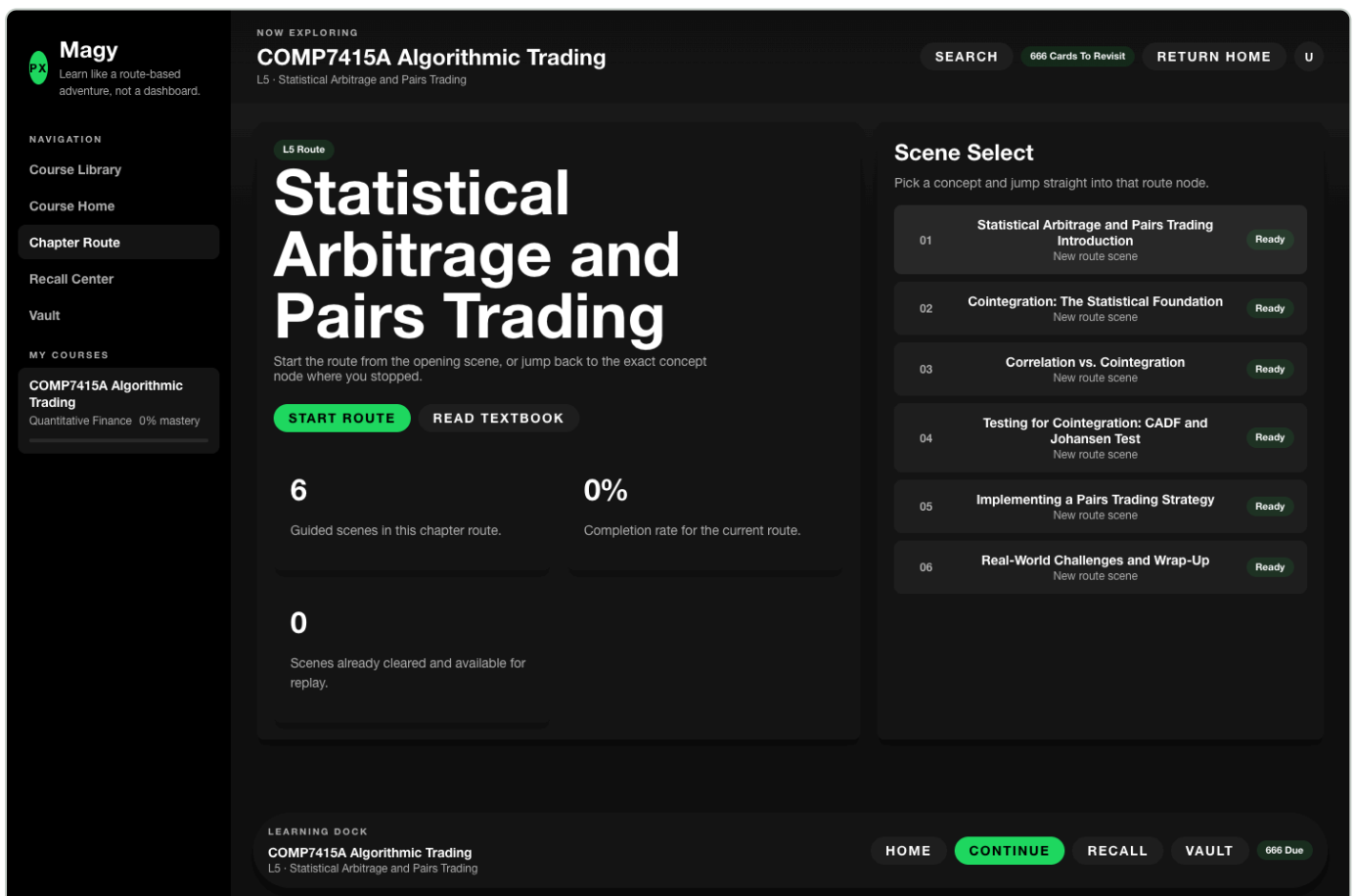


Figure 3. A generated chapter represented as a route with multiple concept scenes.

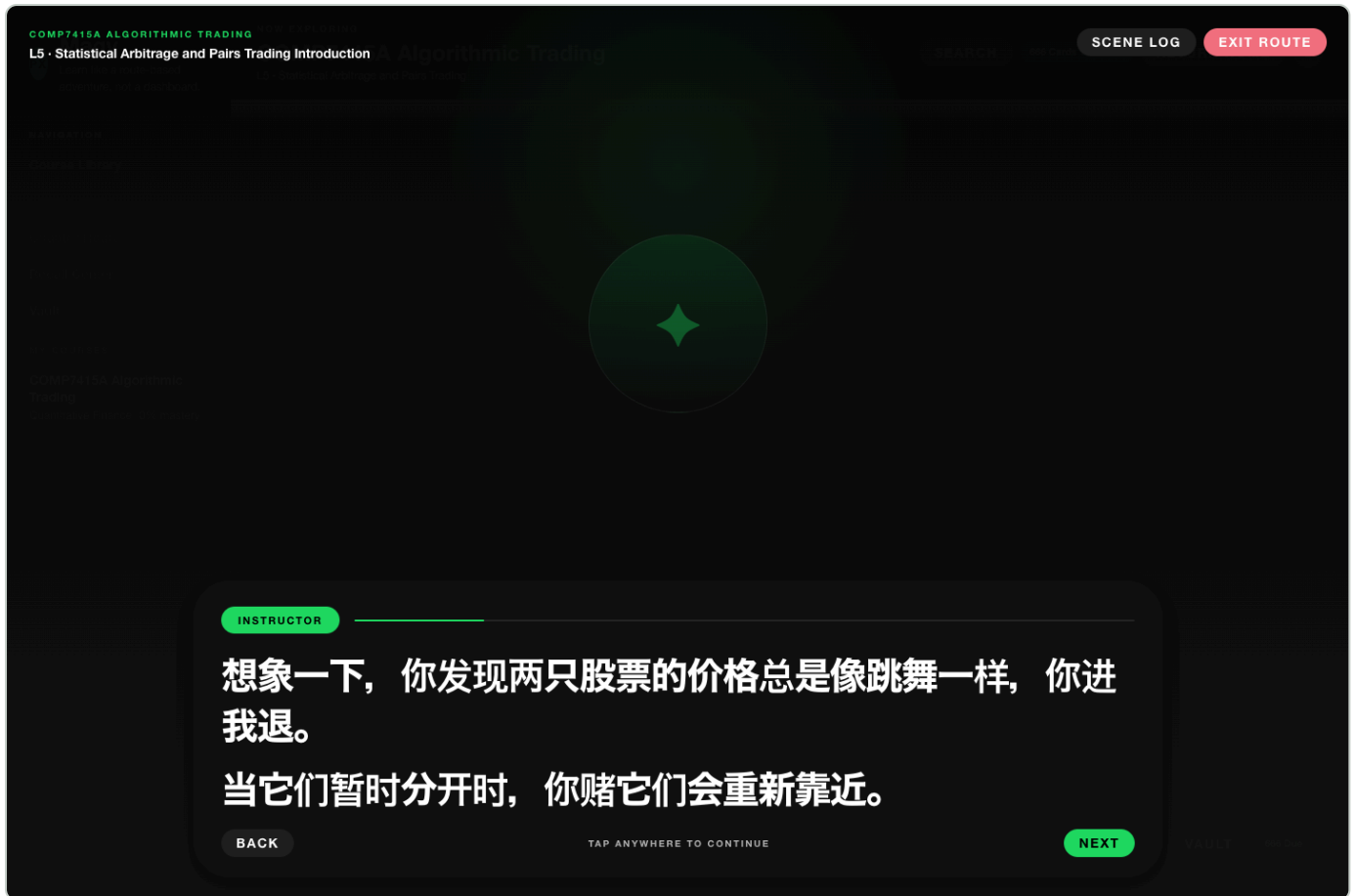


Figure 4. Guided-story playback with a full-screen learning overlay.



Figure 5. Textbook reading mode using generated MDX content.

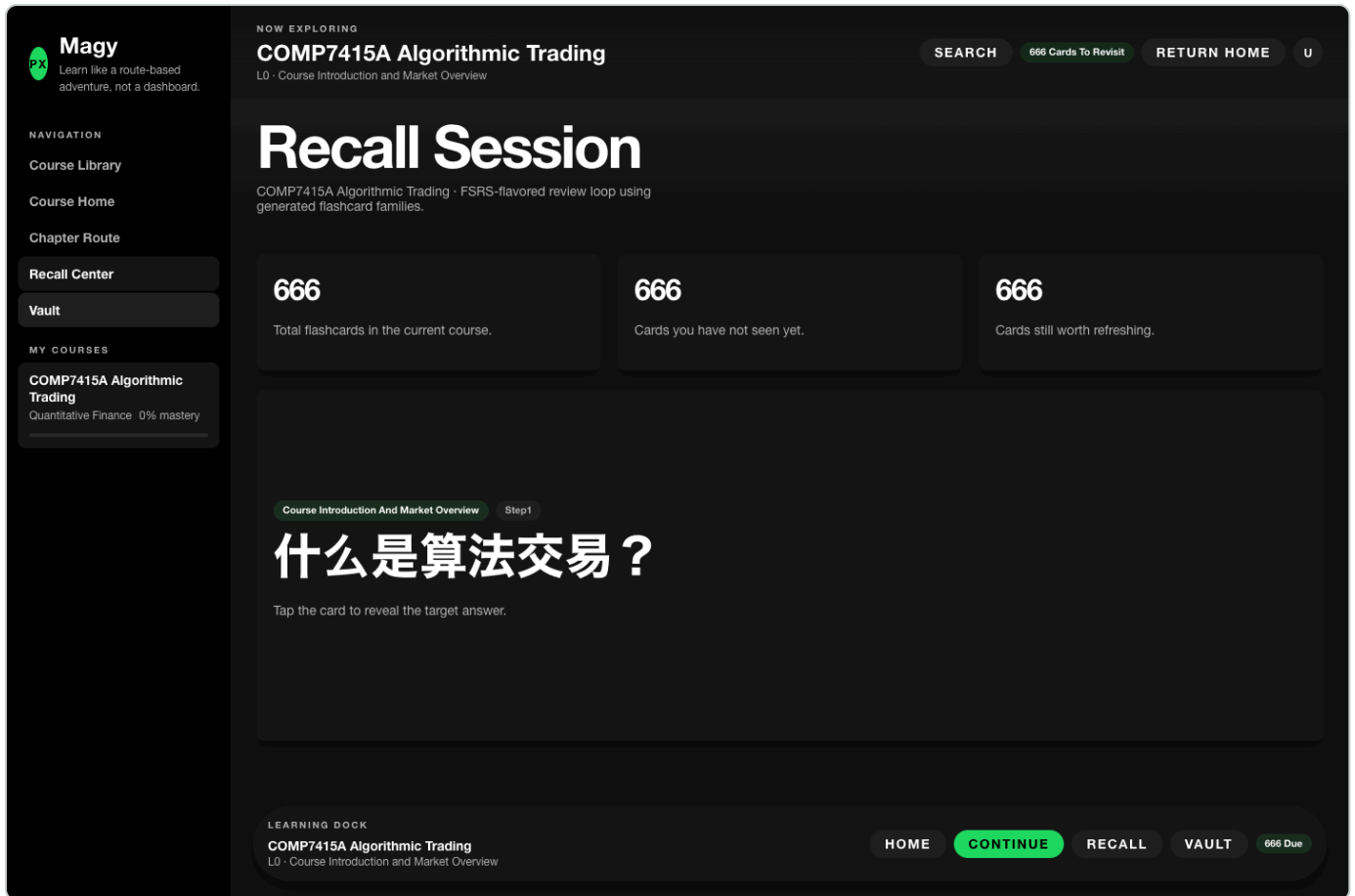


Figure 6. Flashcard-based recall loop using generated practice assets.

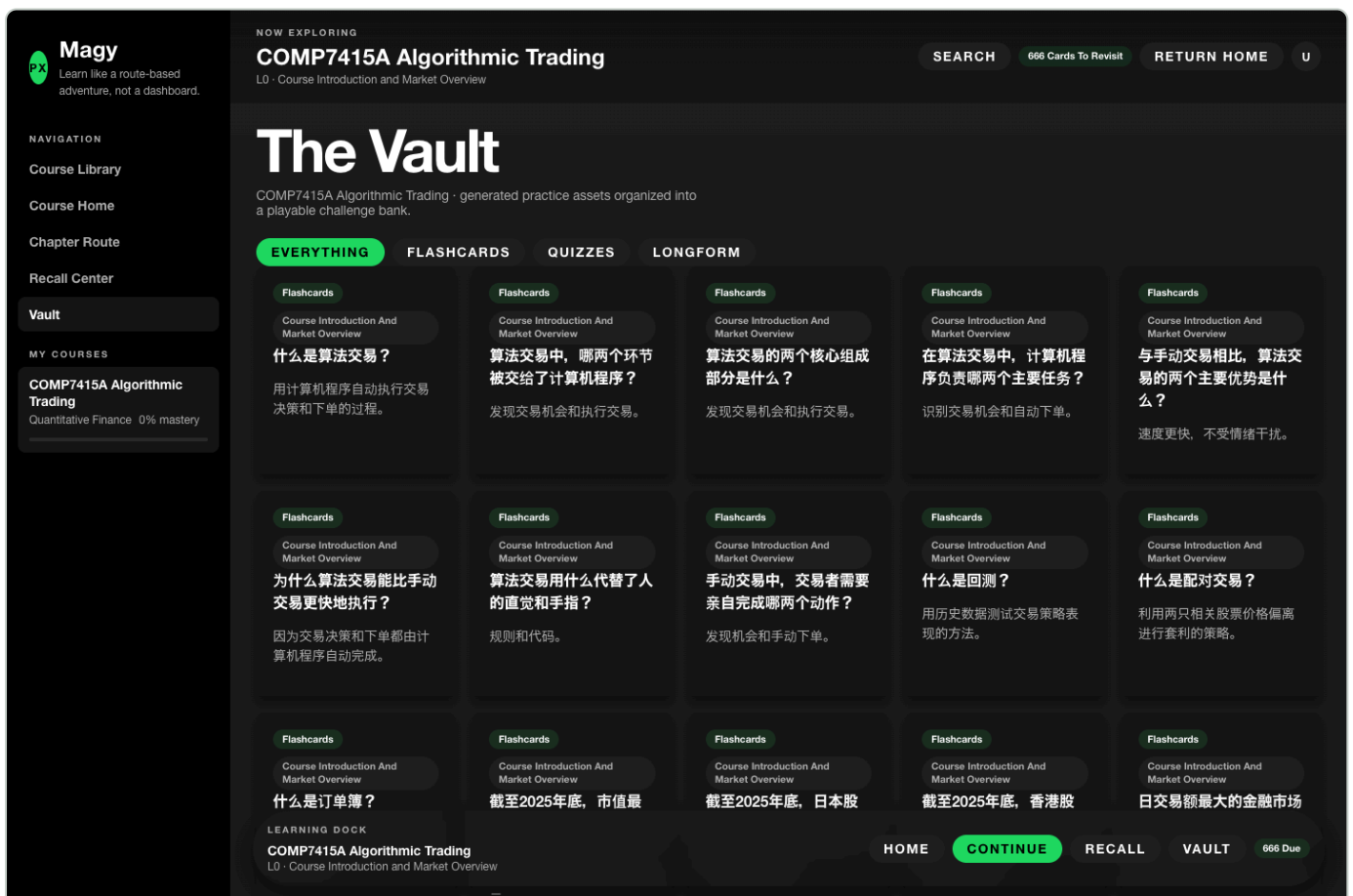


Figure 7. Practice vault separating flashcards, quizzes, and longform questions.

5. Creator Review and Debugging Workspace

In addition to the learner app, the project includes a browser-based creator workspace for inspecting generated artifacts. This workspace is intended for content review and debugging. It is not a full authoring studio yet, but it provides a practical foundation for human-in-the-loop review.

Current creator-side functions include:

- course and lecture selection from the generated catalog;
- overview of primary artifact paths;
- MDX textbook preview;
- guided-story screen inspection;
- step-local question bank inspection;
- separate display of flashcards, quiz items, and longform questions;
- relevance report viewing when sidecar scoring output is available.

This helps address a central risk in AI-generated education: outputs must be reviewable. The creator workspace makes it possible to inspect exactly what the model generated and how it is stored.

Creator Workspace Screenshots

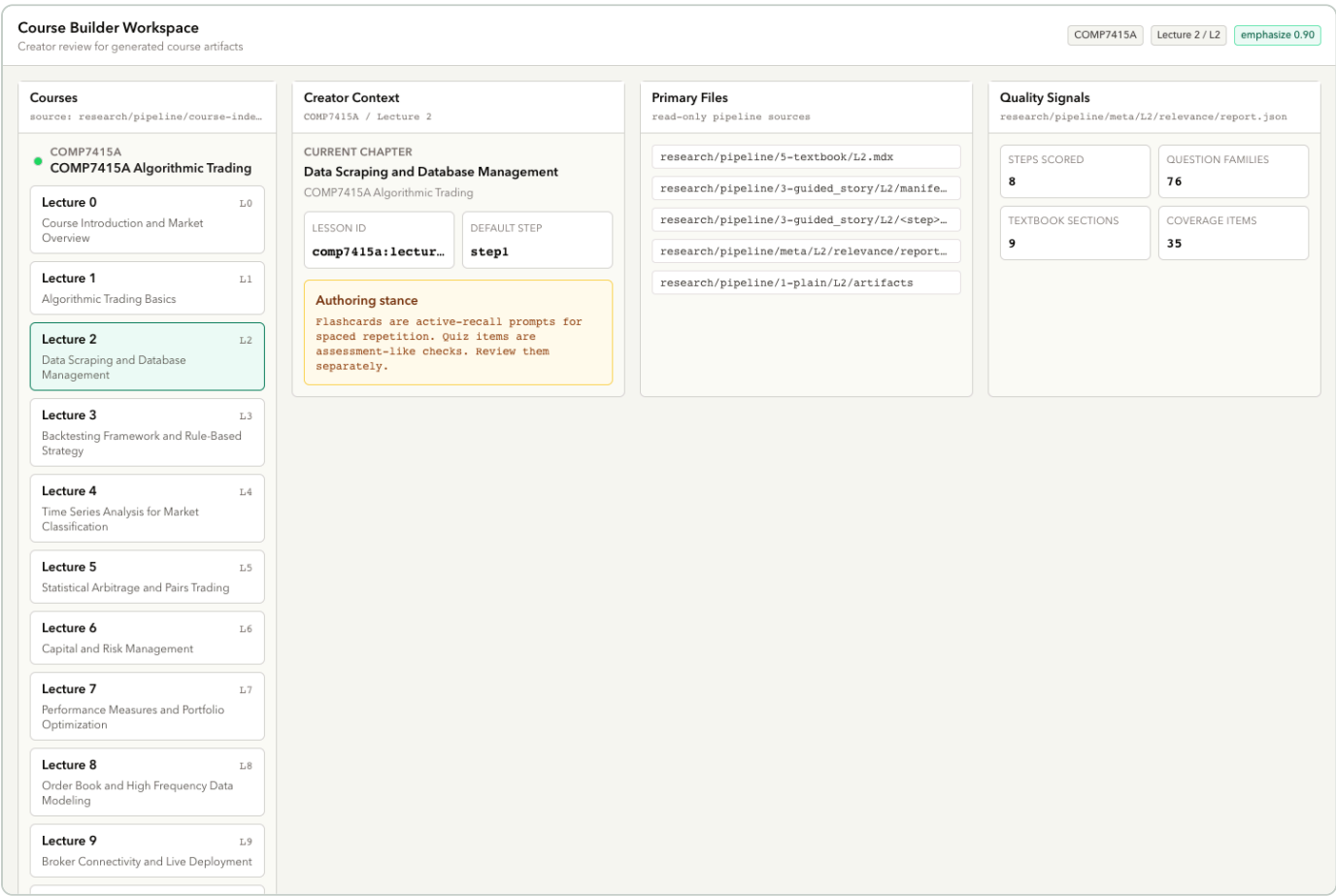


Figure 8. Creator overview for a generated lecture package.

Course Builder Workspace

Creator review for generated course artifacts

COMP7415A

Lecture 2 / L2

emphasize 0.90

Courses

source: research/pipeline/course-inde...

COMP7415A

COMP7415A Algorithmic Trading

Lecture 0

Course Introduction and Market Overview

L.0

Lecture 1

Algorithmic Trading Basics

L.1

Lecture 2

Data Scraping and Database Management

L.2

Lecture 3

Backtesting Framework and Rule-Based Strategy

L.3

Lecture 4

Time Series Analysis for Market Classification

L.4

Lecture 5

Statistical Arbitrage and Pairs Trading

L.5

Lecture 6

Capital and Risk Management

L.6

Lecture 7

Performance Measures and Portfolio Optimization

L.7

Lecture 8

Order Book and High Frequency Data Modeling

L.8

Lecture 9

Broker Connectivity and Live Deployment

L.9

Textbook preview

research/pipeline/5-textbook/L2.mdx

数据爬取与数据库管理

为什么这一课重要

想象一下，你每天需要手动查看几百只股票的价格、新闻和财报。这几乎不可能，而且很容易出错。

现在，让程序替你完成这些重复的工作。这就是 **网络爬取** `term:web_scraping / Web Scraping` 的核心思想。

在 **算法交易** `term:algorithmic_trading / Algorithmic Trading` 中，数据就是燃料。没有数据，策略就是空谈。爬取的数据可以用于：

- 实时数据收集**：自动获取股价、经济指标等。
- 市场情绪分析**：爬取新闻、论坛和社交媒体内容。
- 公司分析**：自动收集公司公告、财报、管理层变动等信息。
- 另类数据**：收集天气、网络流量等非传统金融数据。

KEY POINT

网络爬取是算法交易数据 pipeline 的第一步，它让你能够自动化地、高效地从互联网获取结构化或非结构化数据，为后续的策略开发、回测和分析提供燃料。

Quiz

medium

q_quiz_web_scraping_use_case_v1

PROMPT

以下哪种情况最适合使用网络爬取？

1 手动计算股票平均价格

2 自动从财经网站收集每日收盘价

Textbook relevance

section badges

section_2

emphasize 0.90

importance:0.90 course:0.90 exam:0.50

为什么这一课重要 (#why-this-lesson-matters)

核心动机：解释网络爬取在算法交易中的价值，建立学习驱动力。

section_3

emphasize 0.85

importance:0.85 course:0.85 exam:0.50

课程全景与关键问题 (#concept-map)

课程地图：提供本课的整体框架和关键问题，帮助学生建立知识结构。

section_4

emphasize 0.90

importance:0.90 course:0.90 exam:0.50

网络爬取基础：从网页结构到数据提取 (#web-scraping-foundations)

核心内容：涵盖网页结构和BeautifulSoup使用，是爬取的基础。

section_13

emphasize 0.90

importance:0.90 course:0.90 exam:0.50

实战爬取与可靠数据源 (#real-world-scraping)

核心应用：实战爬取S&P 500和使用yfinance，直接关联课程目标。

section_20

emphasize 0.85

importance:0.85 course:0.85 exam:0.50

数据库管理：存储与查询金融数据 (#database-management)

Figure 9. Generated textbook preview with custom course components.

Course Builder Workspace

Creator review for generated course artifacts

COMP7415A

Lecture 2 / L2

emphasize 0.90

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source: research/pipeline/course-inde...

COMP7415A

COMP7415A Algorithmic Trading

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L.7

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L.8

Lecture 9

Broker Connectivity and Live Deployment

L.9

Steps

research/pipeline/3-guided_story/L2...

step1

What is web scraping and why it matters for algorithmic trading

emphasize

step2

Understanding the basic structure of a website: HTML, CSS, and JavaScript

emphasize

step3

Using BeautifulSoup to parse HTML and extract data

emphasize

step4

Real-world scraping: S&P 500 list and handling anti-bot measures

emphasize

step5

Using yfinance for reliable financial data access

emphasize

step6

Storing scraped data in a SQLite database

emphasize

step7

Database design considerations for financial data

emphasize

step8

Limitations of web scraping and alternative tools

condense

Screens - step1

research/pipeline/3-guided_story/L2/step1/step.json

1.s001

narration

想象一下，你每天需要手动查看几百只股票的价格、新闻和财报。

这几乎不可能，而且很容易出错。

2.s002

narration

web_scraping

现在，让程序替你完成这些重复的工作。

这就是 **网络爬取** `term:web_scraping` 的核心思想。

Introduces web_scraping

3.s003

narration

algorithmic_trading

在 **算法交易** `term:algorithmic_trading` 中，数据就是燃料。

没有数据，策略就是空谈。

Introduces algorithmic_trading

4.s004

narration

爬取的数据可以用于：

实时获取股价、经济指标，分析市场情绪，甚至监控天气数据。

5.s005

exercise

下面哪种情况最适合使用网络爬取？

single_choice

下面哪种情况最适合使用网络爬取？

0. 手动计算股票平均价格

1. 自动从财经网站收集每日收盘价

Question bank

research/pipeline/3-guided_story/L2/step1/question_bank.js...

FLASHCARDS (3)

active recall for spaced repetition

qf_flash_web_scraping_def

easy

short_answer

active recall

emphasize 0.90

网络爬取的核心定义：自动化地从网站提取数据。

学生能准确说出网络爬取的定义。

web_scraping_definition

q_flash_web_scraping_def_v1

q_flash_web_scraping_def_v2

qf_flash_web_scraping_use_cases

easy

short_answer

active recall

emphasize 0.85

网络爬取在算法交易中的具体应用场景。

学生能列举网络爬取在算法交易中的至少两个应用场景。

web_scraping_algo_trading_use_cases

q_flash_web_scraping_use_cases_v1

q_flash_web_scraping_use_cases_v2

qf_flash_data_fuel

easy

short_answer

active recall

emphasize 0.80

数据在算法交易中的比喻性描述。

学生能理解数据在算法交易中的核心地位。

data_as_fuel_for_algo_trading

q_flash_data_fuel_v1

q_flash_data_fuel_v2

Figure 10. Guided-story review surface showing step screens and question bank context.

Course Builder Workspace

Creator review for generated course artifacts

COMP7415A

Lecture 2 / L2

emphasize 0.90

Courses

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L.9

Lecture Steps

research/pipeline/3-guided_story/L2/m...

step1

What is web scraping and why it matters for algorithmic trading

step2

Understanding the basic structure of a website: HTML, CSS, and JavaScript

step3

Using BeautifulSoup to parse HTML and extract data

step4

Real-world scraping: S&P 500 list and handling anti-bot measures

step5

Using yfinance for reliable financial data access

step6

Storing scraped data in a SQLite database

step7

Database design considerations for financial data

step8

Limitations of web scraping and alternative tools

Practice assets - step1

research/pipeline/3-guided_story/L2/step1/question_bank.json

Flashcards are not quiz questions

Review front/back cards for active recall. Keep options and distractors in Quiz, where assessment-style wording belongs.

COVERAGE

web_scraping_definitionweb_scraping_algo_trading_use_casesdata_as_fuel_for_algo_trading

FLASHCARDS (3)

active recall for spaced repetition

q_flash_web_scraping_def

easy

short_answer

active recall

emphasize 0.90

网络爬取的核心定义：自动化地从网站提取数据。

学生能准确说出网络爬取的定义。

web_scraping_definition

q_flash_web_scraping_def_v1

q_flash_web_scraping_def_v2

q_flash_web_scraping_use_cases

easy

short_answer

active recall

emphasize 0.85

网络爬取在算法交易中的具体应用场景。

学生能列举网络爬取在算法交易中的至少两个应用场景。

web_scraping_algo_trading_use_cases

q_flash_web_scraping_use_cases_v1

q_flash_web_scraping_use_cases_v2

q_flash_data_fuel

easy

short_answer

active recall

emphasize 0.80

数据在算法交易中的比喻性描述。

学生能理解数据在算法交易中的核心地位。

data_as_fuel_for_algo_trading

Figure 11. Practice review surface separating flashcards, quiz, and longform items.

Course Builder Workspace

Creator review for generated course artifacts

COMP7415A

Lecture 2 / L2

emphasize 0.90

Courses

source: research/pipeline/course-inde...

COMP7415A

COMP7415A Algorithmic Trading

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L.8

Lecture 9

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L.9

Relevance report

research/pipeline/meta/L2/relevance/report.json

STEPS

step1

emphasize 0.90

importance:0.90

course:0.90

exam:0.50

核心概念：定义网络爬取及其在算法交易中的价值。课程描述明确包含“数据爬取”，是后续所有步骤的基础。

step2

emphasize 0.80

importance:0.80

course:0.80

exam:0.50

基础：理解网页结构（HTML/CSS/JavaScript）是爬取的前提。课程描述包含“数据爬取”，但此步骤偏重技术基础，重要性略低于step1。

step3

emphasize 0.90

importance:0.90

course:0.90

exam:0.50

核心技能：使用BeautifulSoup解析HTML和提取数据。这是实现“数据爬取”的关键工具，课程描述明确包含。

step4

emphasize 0.85

importance:0.85

course:0.85

exam:0.50

实战应用：爬取S&P 500列表，处理反爬机制（User-Agent）。将理论应用于实际金融数据，课程描述包含“数据爬取”。

step5

emphasize 0.90

importance:0.90

course:0.90

exam:0.50

核心工具与概念：使用yfinance获取可靠金融数据，理解调整收盘价。课程描述包含“数据爬取”和“数据库管理”，调整收盘价是金融数据的关键概念。

step6

emphasize 0.85

importance:0.85

course:0.85

exam:0.50

核心技能：将爬取的数据存入SQLite数据库，学习SQL CRUD操作。课程描述明确包含“数据库管理”。

step7

emphasize 0.80

importance:0.80

course:0.80

exam:0.50

高级设计：数据库设计权衡（大表问题、分区策略）。课程描述包含“数据库管理”，是构建可扩展数据管道的关键。

steps

condense 0.70

importance:0.70

course:0.70

exam:0.50

补充知识：爬取的局限性和替代工具（Selenium,XPath）。课程描述包含“数据爬取”，但此步

Coverage and textbook

coverage_scores / textbook_section_scores

COVERAGE

web_scraping_definition

emphasize 0.90

importance:0.90

course:0.90

exam:0.50

网络爬取的定义：从网站自动提取数据的过程。

核心概念：网络爬取的定义，是课程“数据爬取”部分的基础。

web_scraping_algo_trading_use_cases

emphasize 0.85

importance:0.85

course:0.85

exam:0.50

网络爬取在算法交易中的常见用例：实时数据收集、市场情绪分析、公司分析、另类数据。

核心应用：网络爬取在算法交易中的用例，直接关联课程目标。

data_as_fuel_for_algo_trading

emphasize 0.80

importance:0.80

course:0.80

exam:0.50

数据是算法交易的燃料，没有数据策略就是空谈。

核心比喻：数据是算法交易的燃料，强调数据的重要性。

basic_web_structure_three_components

emphasize 0.80

importance:0.80

course:0.80

exam:0.50

理解网页由 HTML、CSS 和 JavaScript 三部分组成

基础概念：网页三要素，是理解爬取目标的基础。

html_purpose_and_nature

emphasize 0.80

importance:0.80

course:0.80

exam:0.50

HTML 定义网页的结构和内容，是一种标记语言而非编程语言

核心概念：HTML 的用途和本质，是爬取的主要对象。

css_purpose

condense 0.60

importance:0.60

course:0.60

exam:0.50

CSS 控制网页的视觉呈现和布局

Figure 12. Relevance sidecar report for generated steps, question families, coverage, and textbook sections.

6. Progress Against the Proposal

The original proposal described four major components: an AI-assisted generation pipeline, a web-based authoring and review environment, a structured course editor, and a learner-facing mobile application.

The current implementation has achieved the first and fourth components as working prototypes, and has partially achieved the second and third components through the creator workspace.

Original objective	Current progress
Structured representation for modular learning content	Implemented through course index, guided-story manifests, step-local <code>step.json</code> , step-local <code>question_bank.json</code> , and MDX textbooks
AI-assisted pipeline	Implemented through <code>coursegen run</code> , conversion, generation, relevance scoring, and provider configuration
Review and validation workflow	Partially implemented through creator workspace, relevance reports, and backend catalog validation
Learner-facing gamified delivery	Implemented as a React player with library, routes, story mode, recall, vault, and textbook mode
Evaluation of feasibility	Initial feasibility demonstrated through one complete 11-chapter generated course package

The implementation has therefore progressed beyond a single isolated AI prompt. It now has an end-to-end path from course material to structured assets to learner-facing interaction.

7. Technical Highlights

The most important technical progress so far is the decision to make generated content structured and validated.

Key engineering decisions include:

- Step-local ownership: each concept step owns its own guided-story content and question bank.
- Single course catalog: the player reads from `course-index.json` instead of inventing a second frontend-only catalog.
- Backend validation: catalog readiness is determined by the backend and then consumed by the frontend.
- Resume-friendly generation: existing generated assets are reused unless explicitly forced.
- Auditability: prompts, requests, raw responses, and content outputs are saved under pipeline metadata.
- Provider abstraction: DeepSeek is treated as an explicit API provider for large-context generation use cases.

These decisions reduce duplicated state, lower AI regeneration cost, and make generated output easier to inspect.

8. Limitations

Several limitations remain:

- The system currently demonstrates one course domain, algorithmic trading, rather than many unrelated subjects.
- The creator workspace is still primarily a review and debugging surface, not a complete authoring studio.
- Some quality checks are implemented, but deeper pedagogical validation still needs to be expanded.
- The learner app currently runs as a web prototype. Mobile packaging and device-level testing remain future work.
- Generated content still requires review for correctness, phrasing quality, and exam alignment.

These limitations are expected at the current stage. The main prototype risk has shifted from basic feasibility to quality control, authoring workflow, and evaluation.

9. Next Steps

The next stage will focus on strengthening the system from a working prototype into a more complete project demonstration.

Planned work:

1. Expand deterministic validation for generated guided stories, question banks, textbook references, term catalogs, and coverage traces.
2. Improve the creator workflow so reviewers can approve, reject, and edit generated content more directly.
3. Continue improving prompts and gates to reduce low-quality or meta-level questions.
4. Package the learner experience for mobile or mobile-like delivery and test responsive interaction.
5. Add more evaluation evidence, including build checks, validation reports, and sample learner walkthroughs.
6. Prepare the interim presentation around the completed end-to-end workflow.

The immediate target is to show that Magy can support the full workflow proposed at the start of the project: source material ingestion, AI-assisted course generation, structured validation, human review, and interactive learner delivery.

10. Conclusion

Magy currently has a working vertical slice of the proposed platform. The project now demonstrates both sides of the system: content can be generated into structured course assets, and those assets can be consumed by a learner-facing interactive application.

The next phase should focus less on proving that generation and rendering are possible, and more on improving reliability, reviewability, and evaluation quality. This is the right direction for turning the prototype into a strong final computing project.